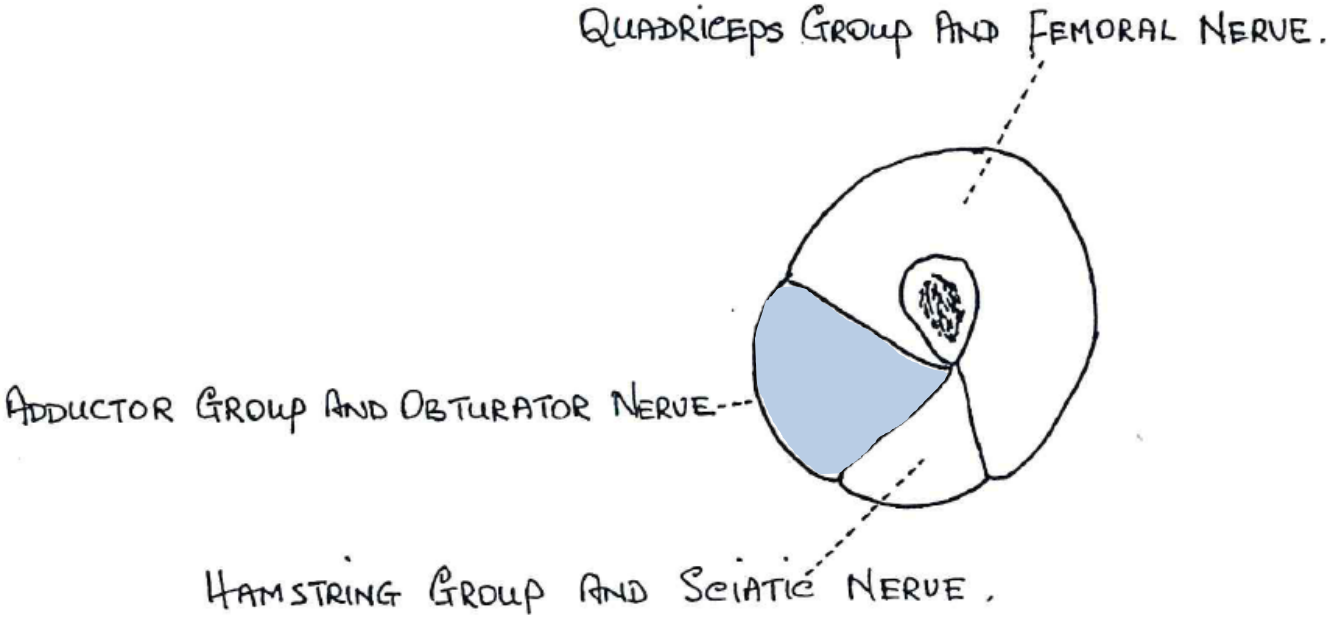
Medial compartment of thigh

Topics

- Muscles of medial compartment
- Femoral factures
- Obturator nerve
- Obturator artery
- Hunter’s canal
- Pes anserinus bursitis
- Corona mortis
  crown of death
- Referred pain hip/knee

Medial Compartment of thigh



Muscles of medial compartment

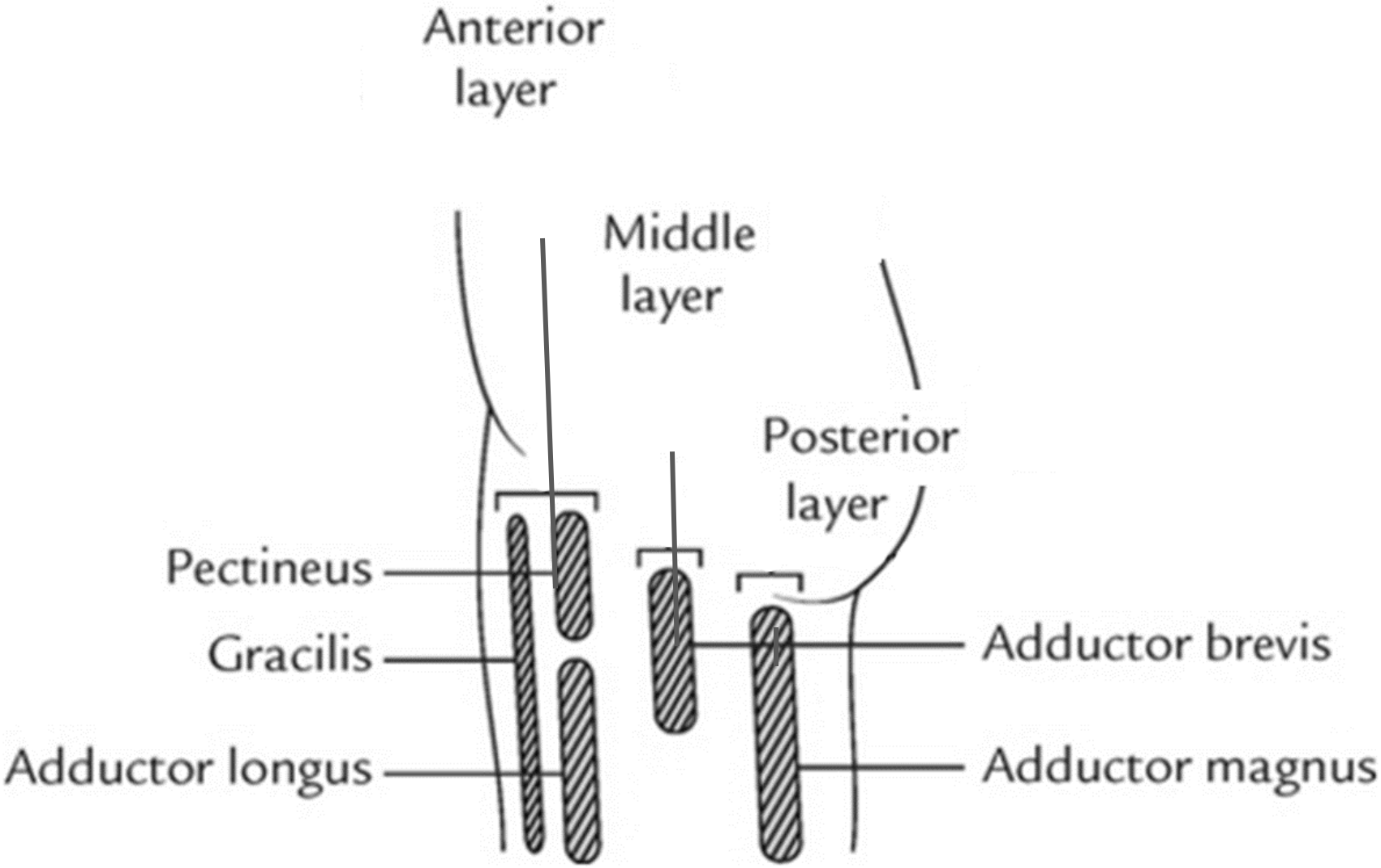
Anterior	→	Adductor longus Pectineus Gracilis
Middle	→	Adductor brevis
Posterior	→	Adductor magnus

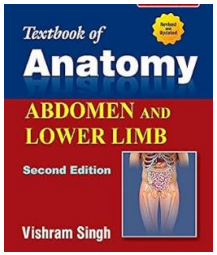
Shared Action

Adduction of thigh

Shared innervation

Obturator nerve


Anterior layer
Middle layer
Posterior layer
Pectineus
Gracilis
Adductor longus
Adductor brevis
Adductor magnus

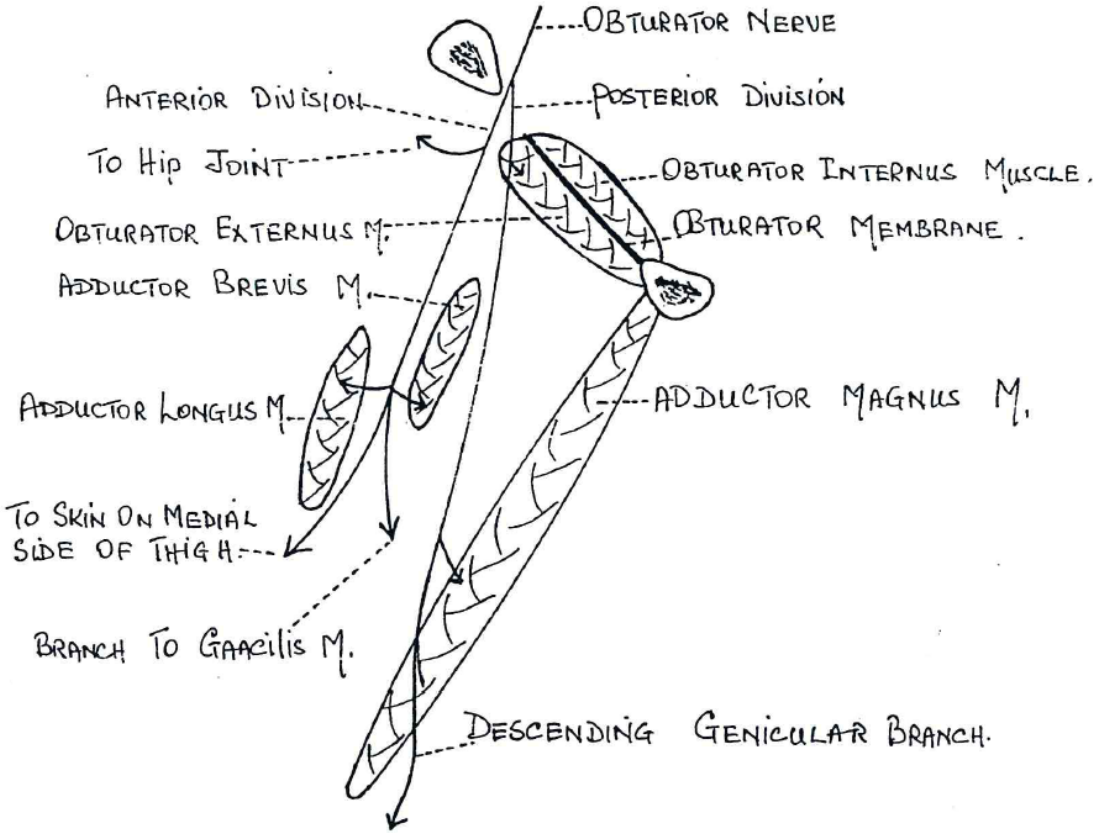

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The distribution of the obturator nerve

travels through obturator foramen and canal

posterior division innervates the posterior group, anterior division innervates the anterior group,
then the middle group is innervated by branches from both posterior and anterior divisions

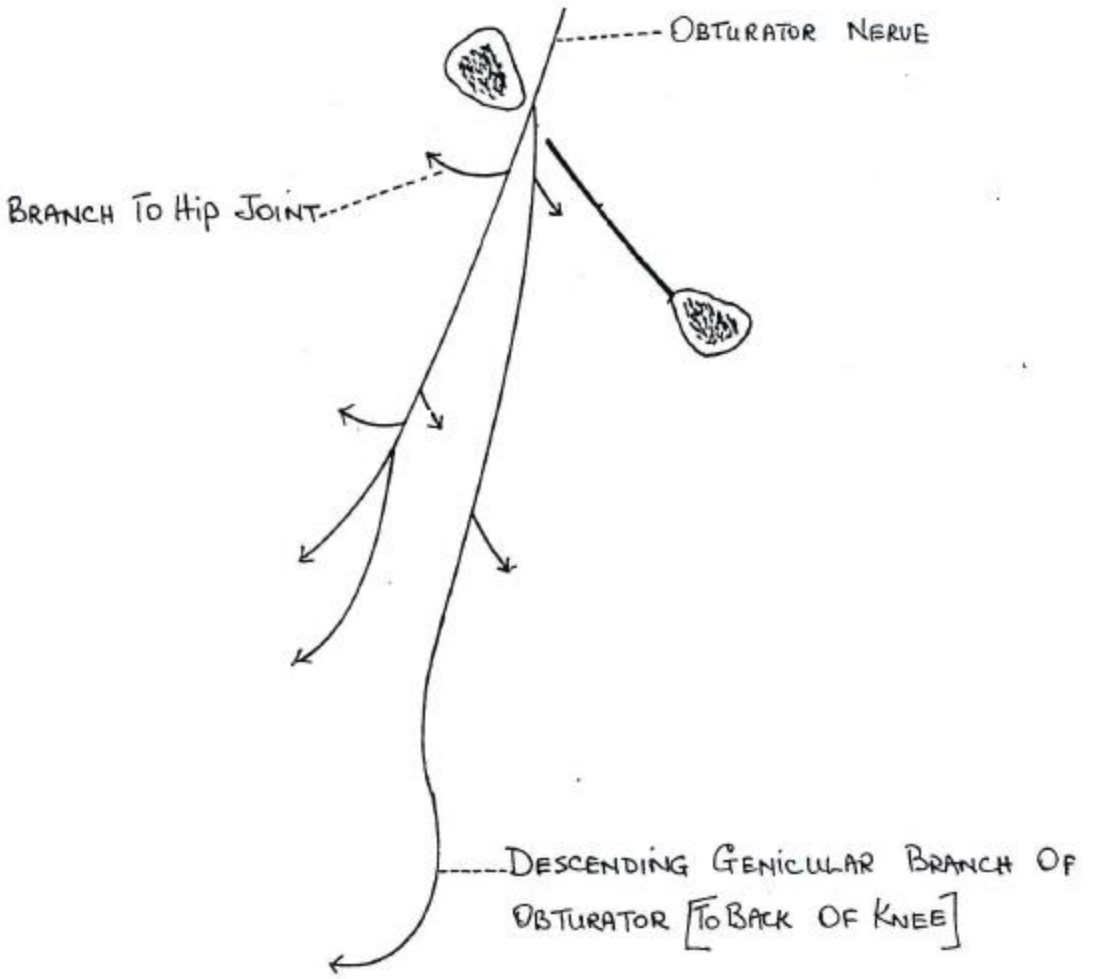
because both the hip and knee joints are innervated by the eobturator nerve, there can be referred pain.
possible because the primary neurons from hip and knee synapse at the same secondary neurone



Coakley Diagram

Pain in the hip and pain in the knee

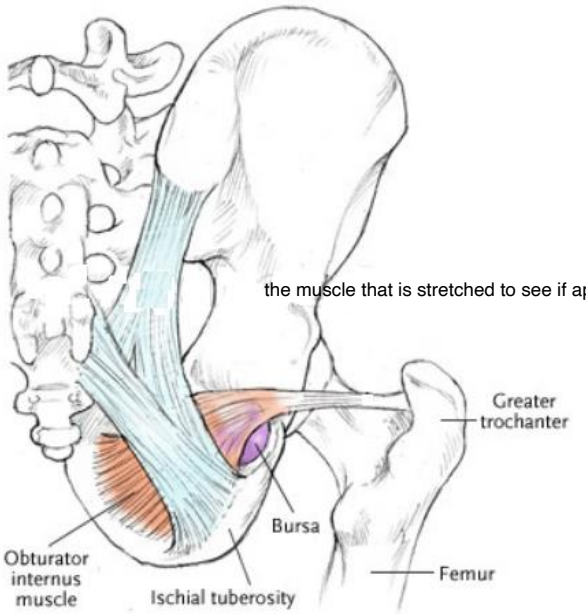
Referred pain L2,3,4
knee joint pain may be referred to hip via obturator (it supplies both joints)



Coakley Diagram

Obturator Internus

both are hip external rotators

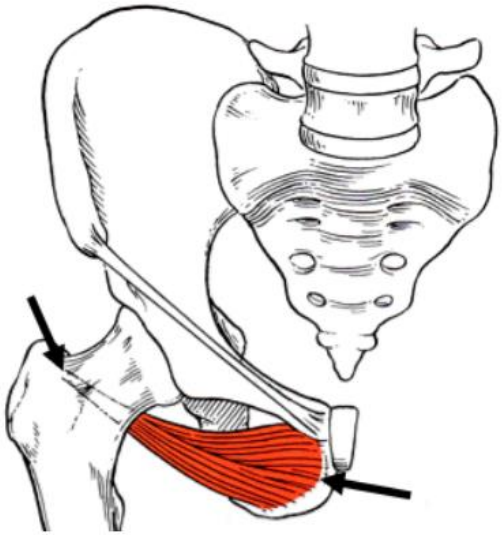


the muscle that is stretched to see if appendix is stuck in there

Posterior view

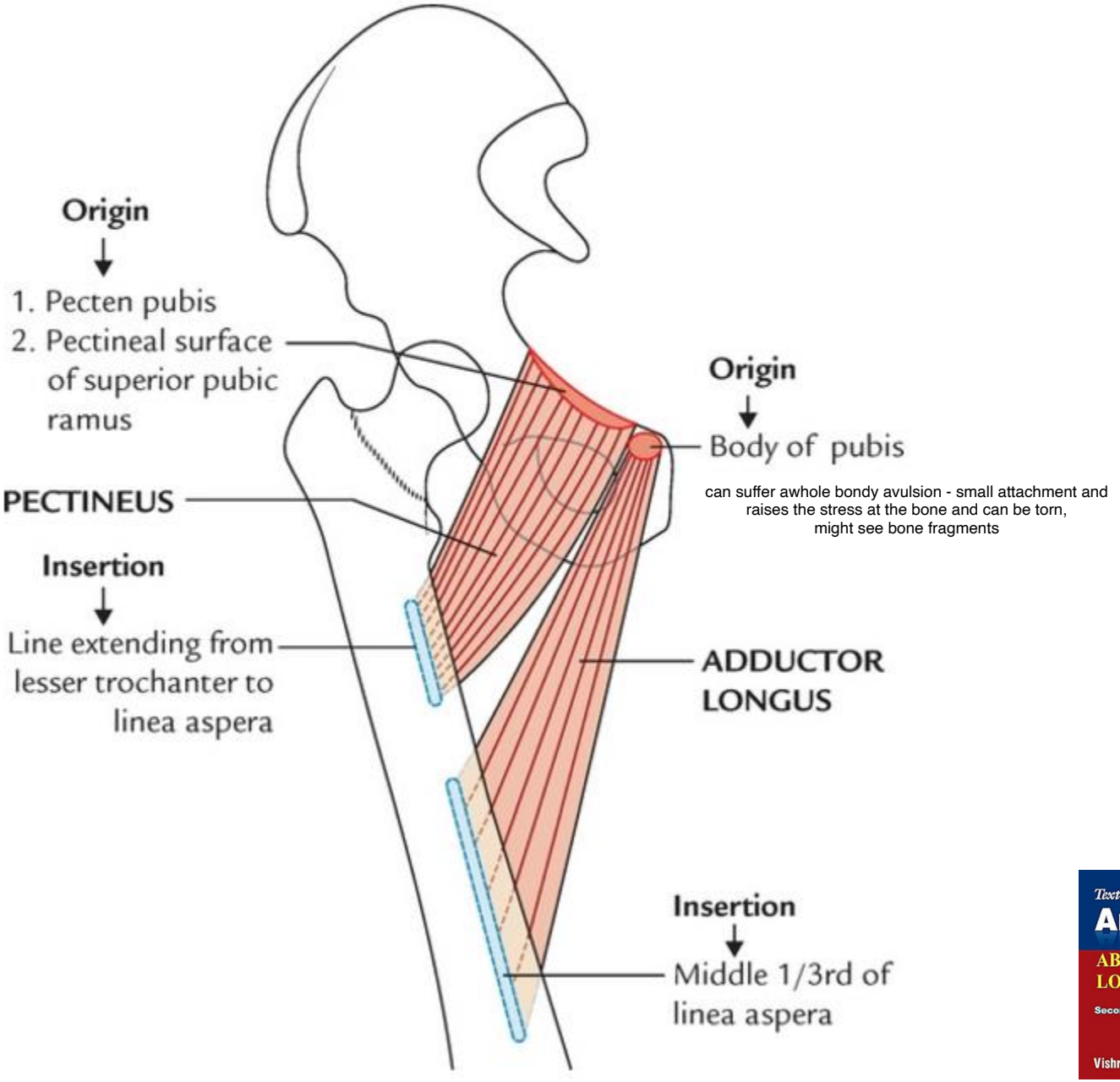
Nerve to obturator internus
Inferior gluteal artery

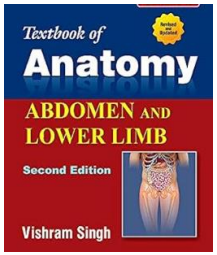
Obturator Externus



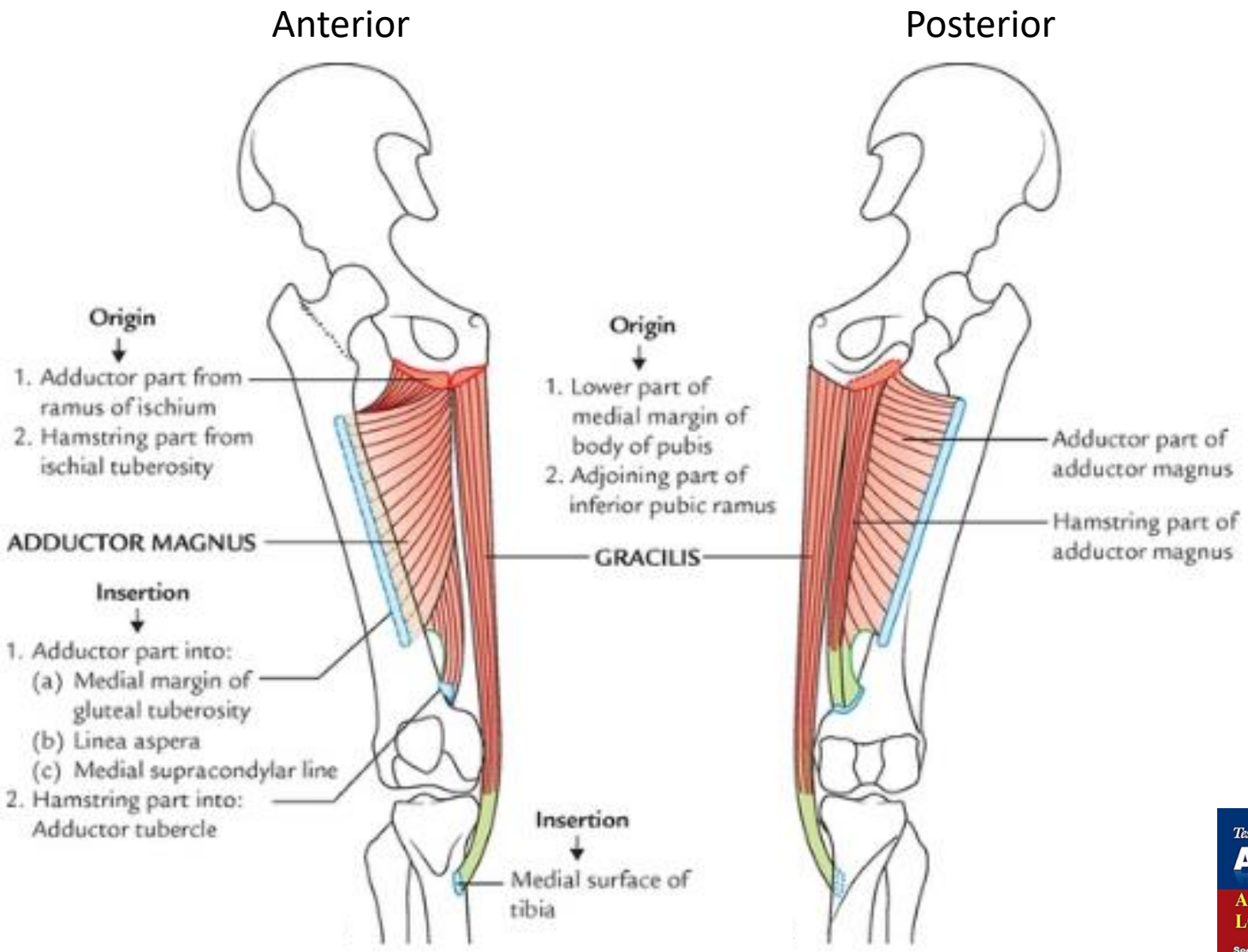
Anterior view

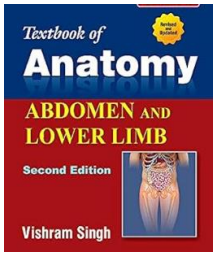
Obturator nerve (post. Div.)
Obturator artery

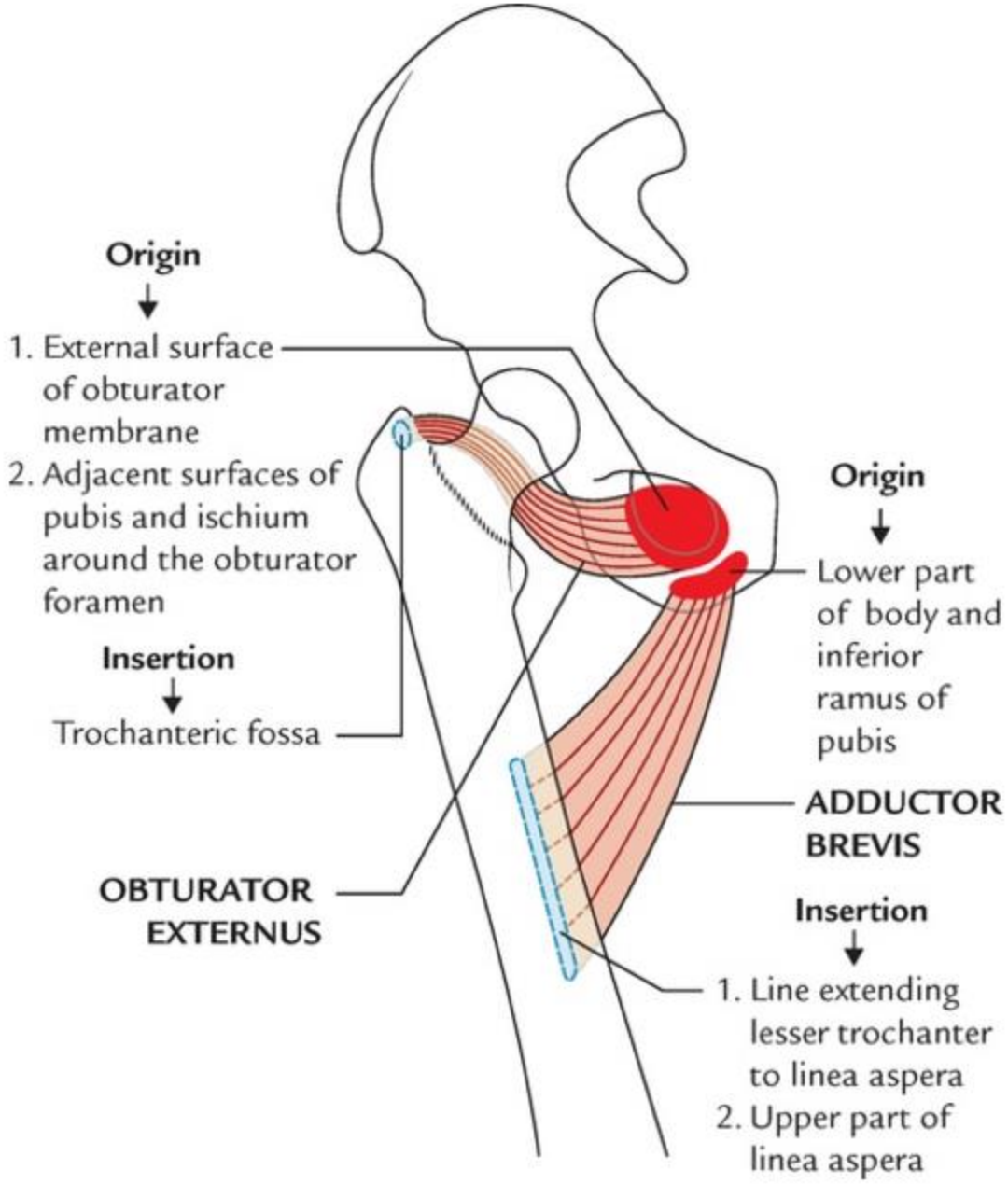

Origin
1. Pecten pubis
2. Pectineal surface of superior pubic ramus
PECTINEUS
Insertion
Line extending from lesser trochanter to linea aspera
Origin
Body of pubis
can suffer awhole bondy avulsion - small attachment and raises the stress at the bone and can be torn, might see bone fragments
ADDUCTOR LONGUS
Insertion
Middle 1/3rd of linea aspera

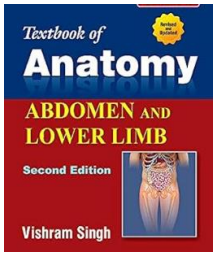

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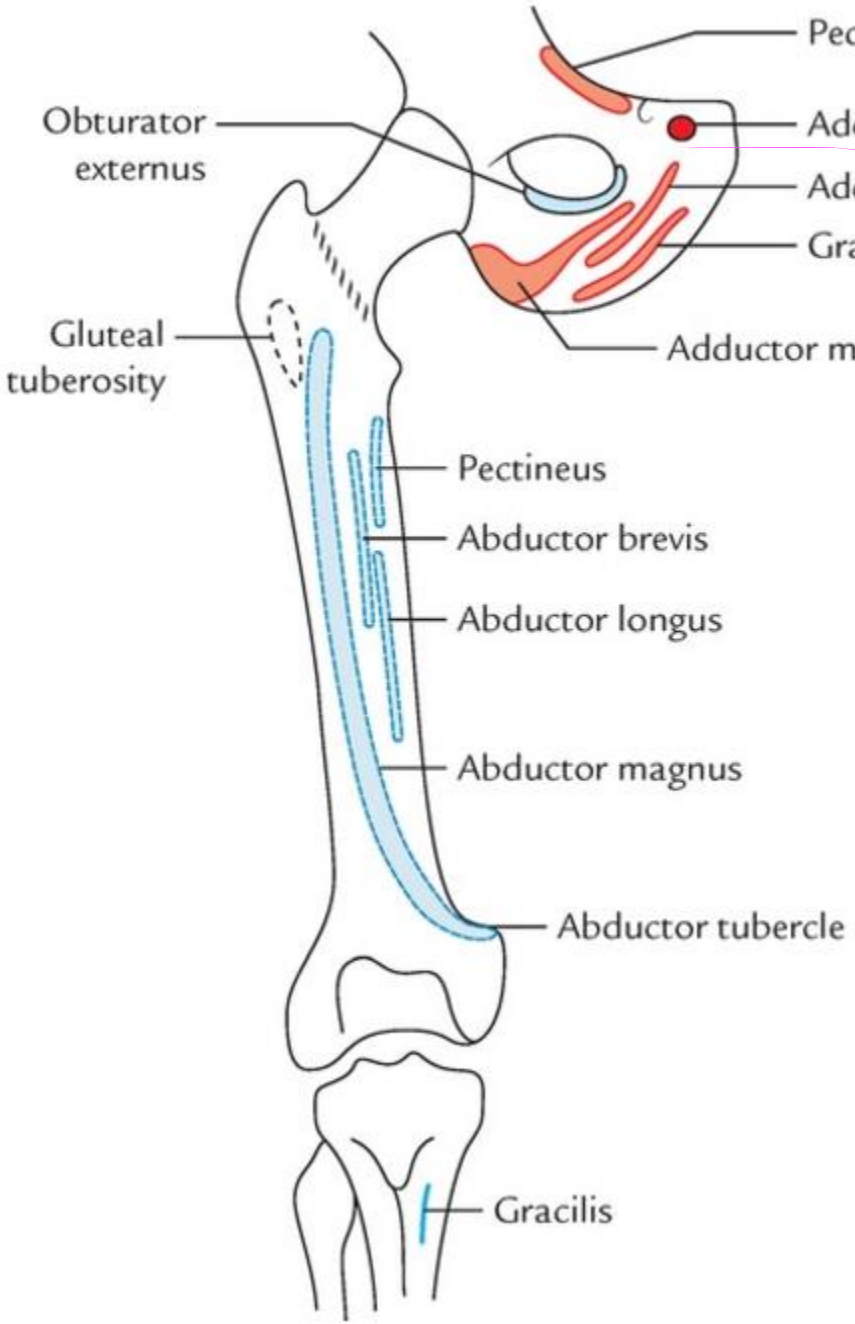
Adductor Magnus and Gracilis



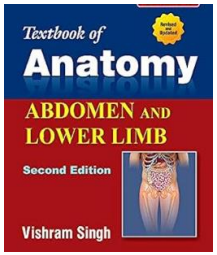



Origin
1. External surface of obturator membrane
2. Adjacent surfaces of pubis and ischium around the obturator foramen
Insertion
Trochanteric fossa
OBTURATOR EXTERNUS
Origin
Lower part of body and inferior ramus of pubis
ADDUCTOR BREVIS
Insertion
1. Line extending lesser trochanter to linea aspera
2. Upper part of linea aspera

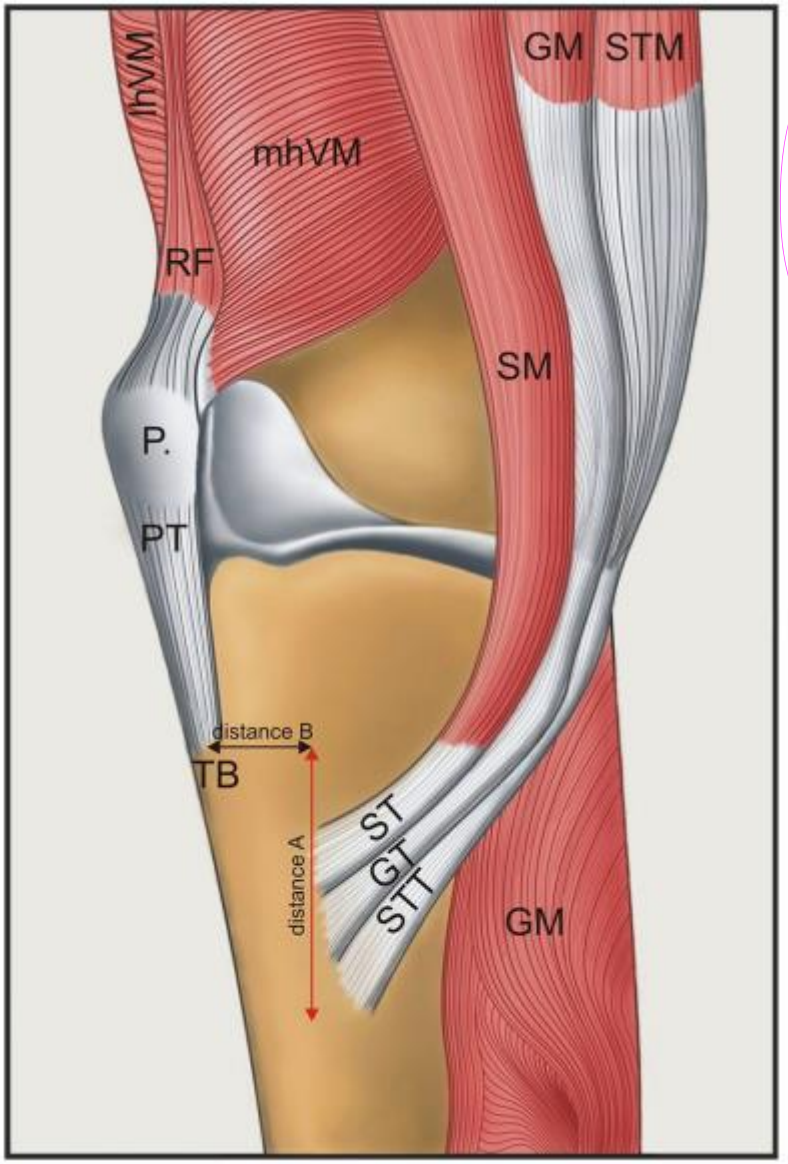

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Pectineus
Adductor longus
Adductor brevis
Gracilis
Adductor magnus
Obturator externus
Gluteal tuberosity
Pectineus
Abductor brevis
Abductor longus
Abductor magnus
Abductor tubercle
Gracilis

can rip off


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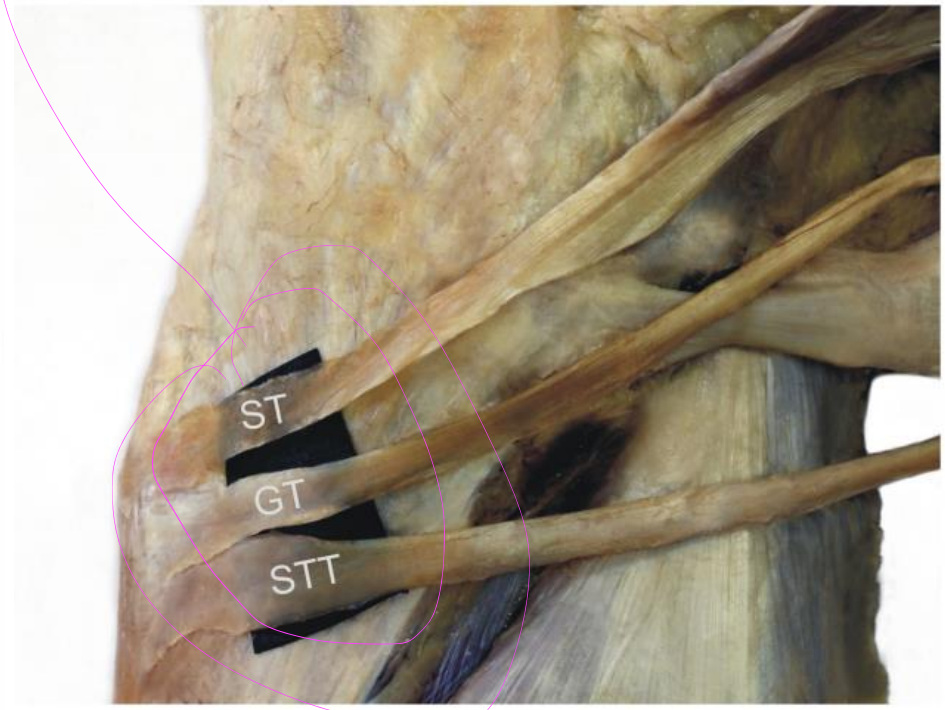
Pes anserinus



"Foot of the goose"



if inflamed then there is a very localised pain



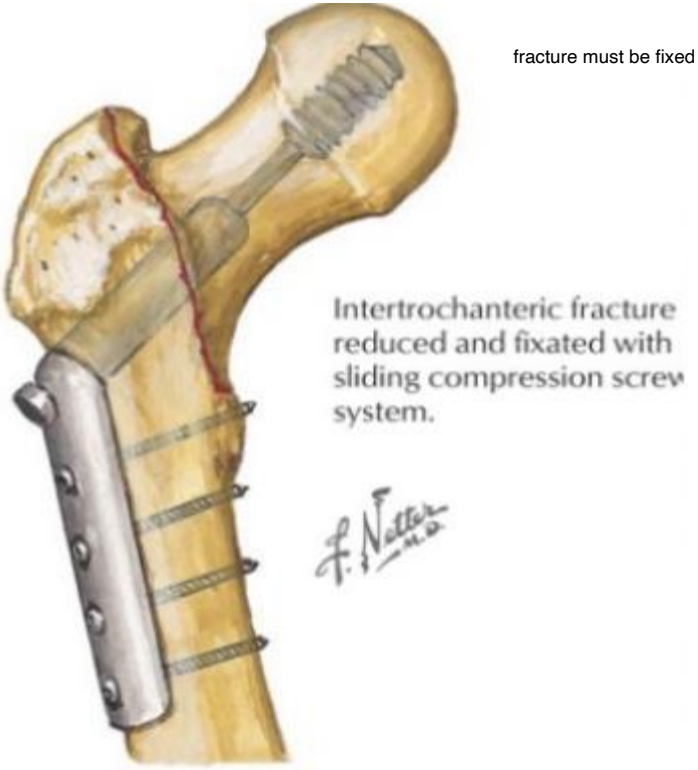
ST sartorius tendon
GT gracilis tendon
STT semitendinosus tendon

A proposal for a new classification of pes anserinus morphology

Łukasz Olewnik[1] · Bartosz Gonera[1] · Michał Podgórski[4] · Michał Polguj[2] · Hubert Jezierski[3] · Mirosław Topol[1]

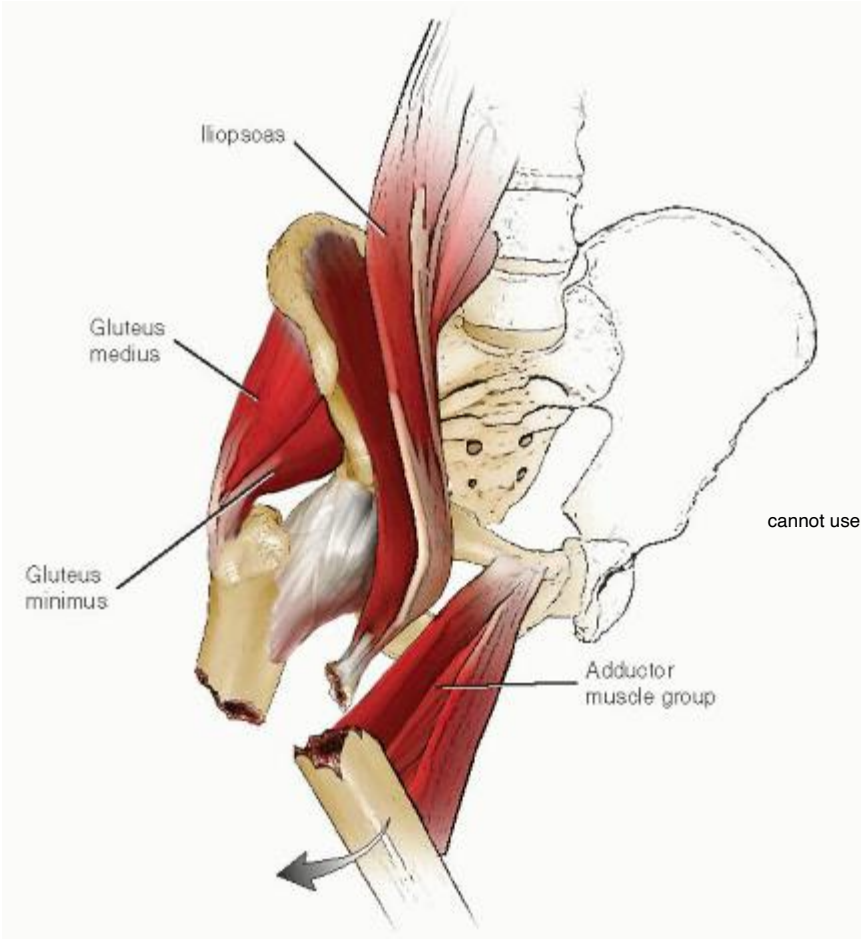
Knee Surgery, Sports Traumatology, Arthroscopy (2019) 27:2984–2993

Intertrochanteric fracture



fracture must be fixed because they have strong muscles connected

Subtrochanteric fracture associated with severe dislocation



cannot use plate because it will bend due to very strong muscles

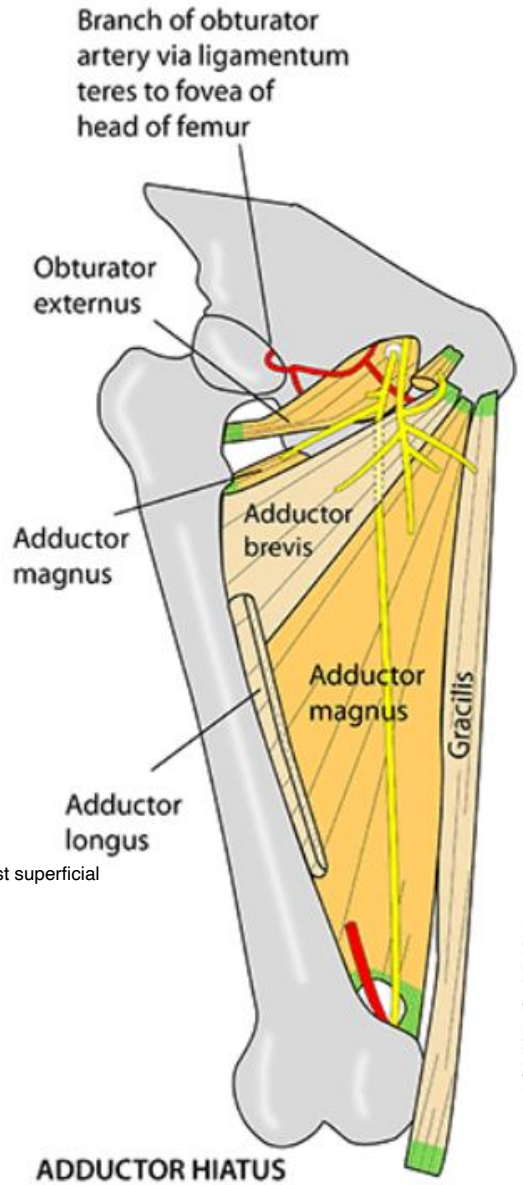
Musculoskeletal Key
Fastest Musculoskeletal Insight Engine

Cephalomedullary nailing

for subtrochanteric fractures



MEDIAL THIGH



most superficial

OBTURATOR NERVE

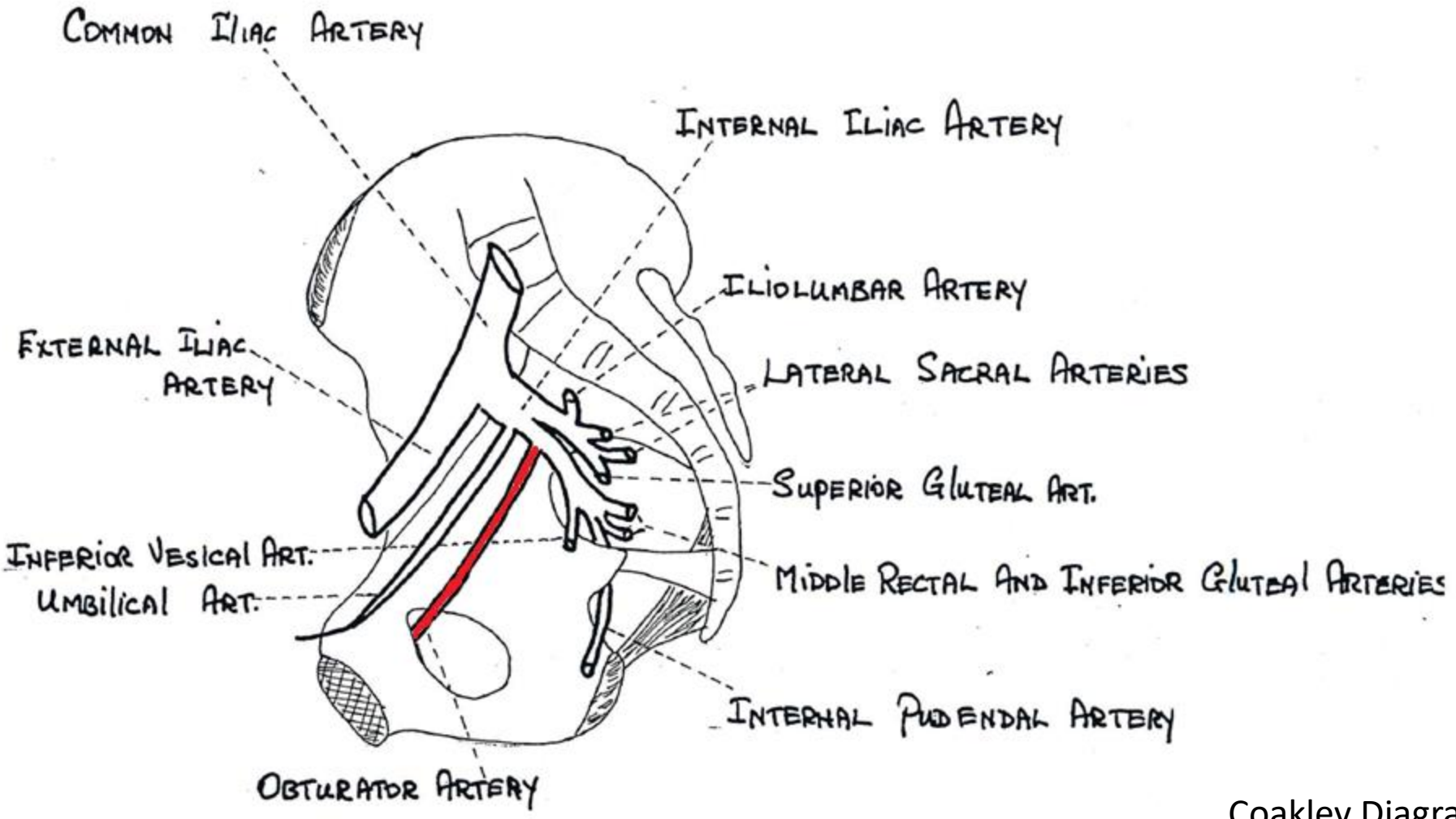
From anterior divisions of L2,3,4

Medial collateral ligament of knee is probably a remnant of the tendon of the hamstring portion of adductor magnus that was originally attached to the tibia

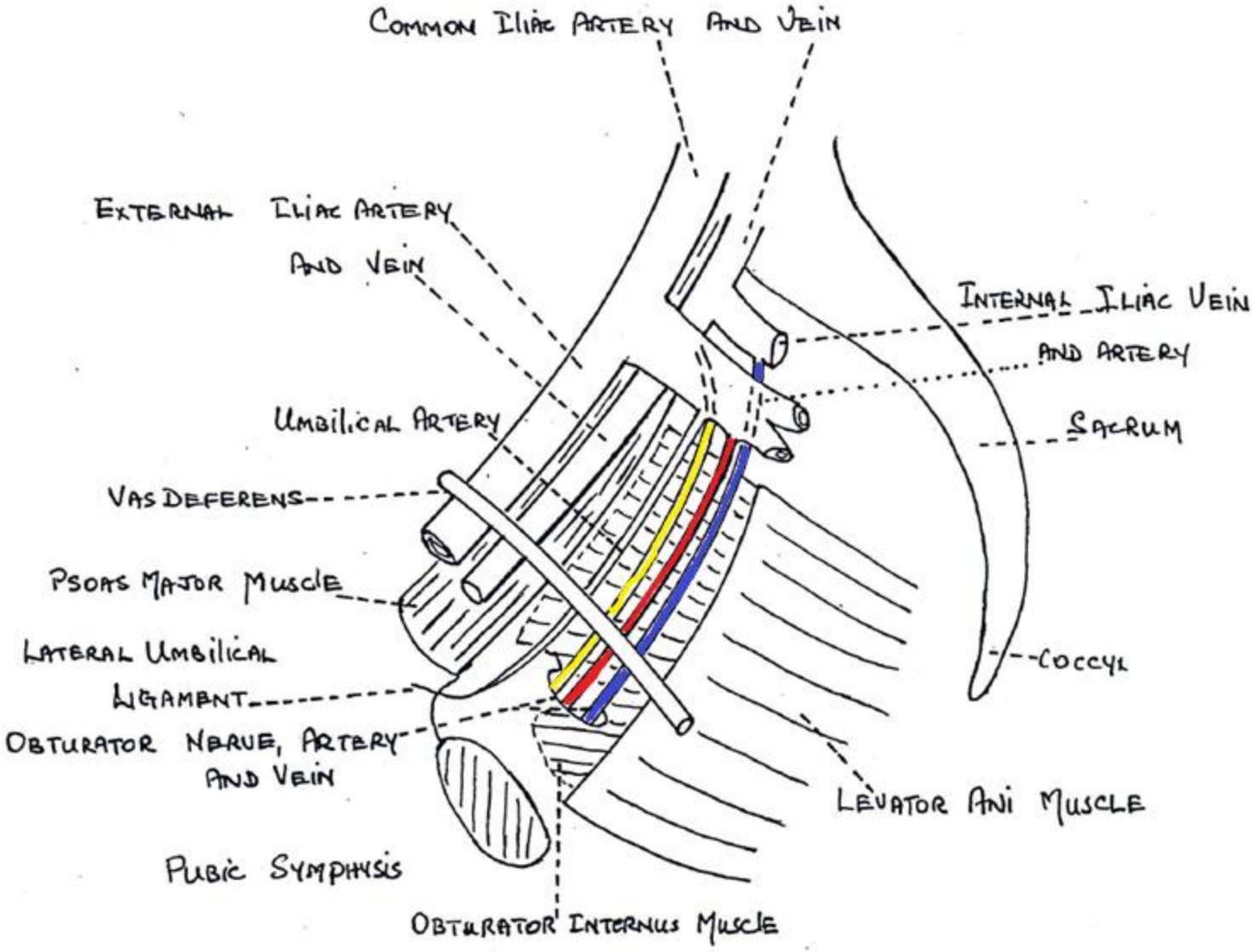
ADDUCTOR HIATUS

Transmits femoral artery, femoral vein, the small genicular branch of the posterior branch of the obturator nerve.

Diagram to show position of internal iliac artery And its two subdivisions in the MALE



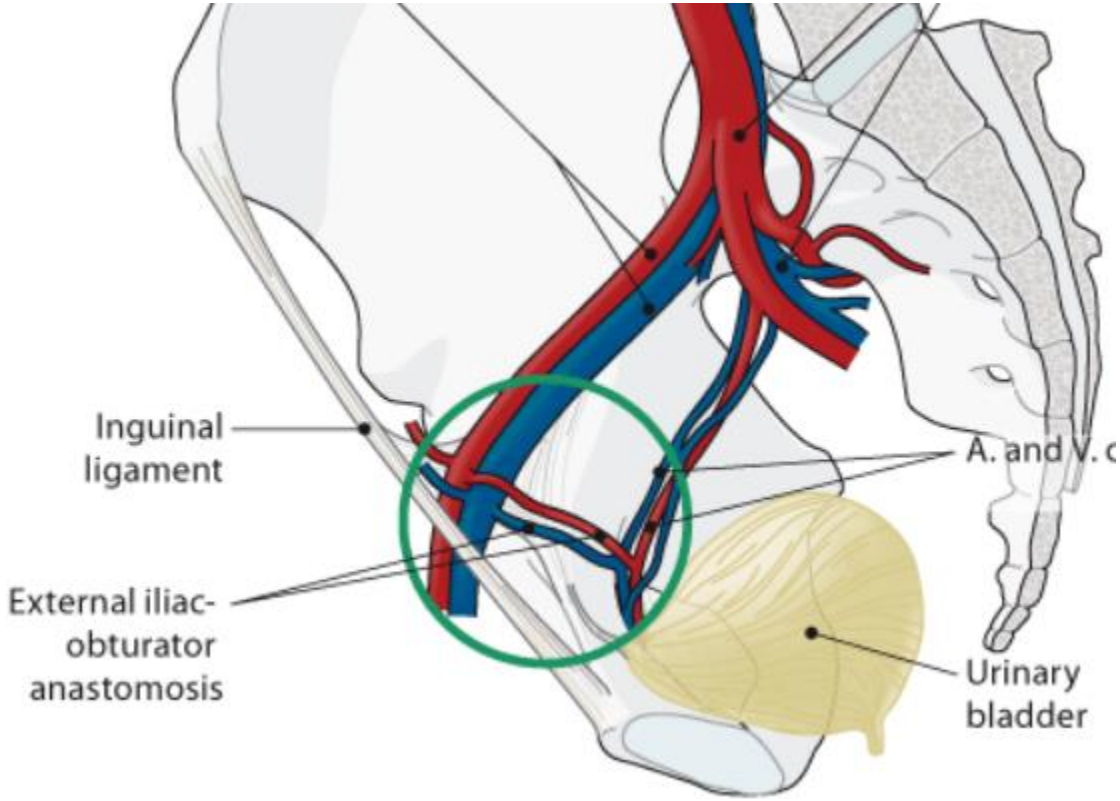
Coakley Diagra



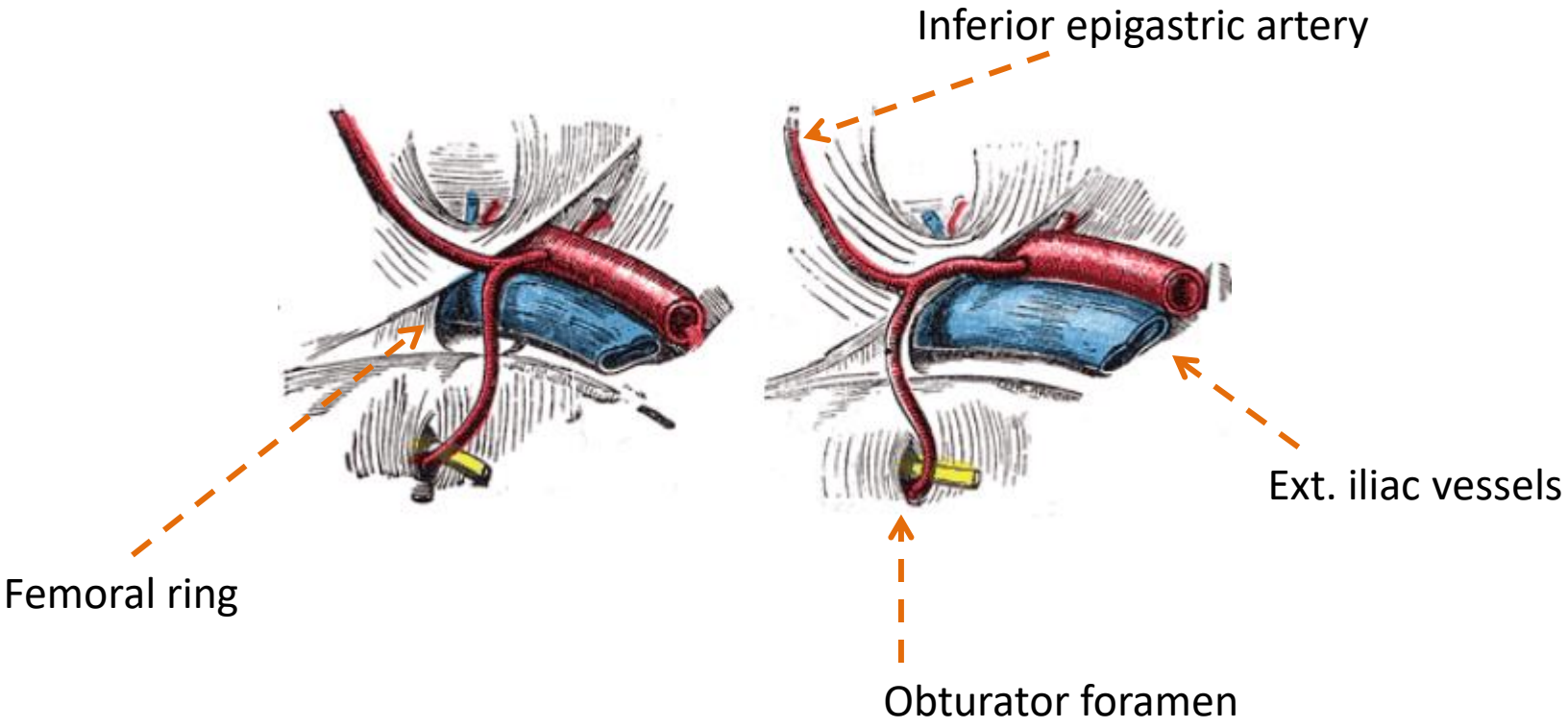
Coakley Diagram

A common variant (33%) of obturator artery communication is the “corona mortis”

communication between the obturator and external iliac

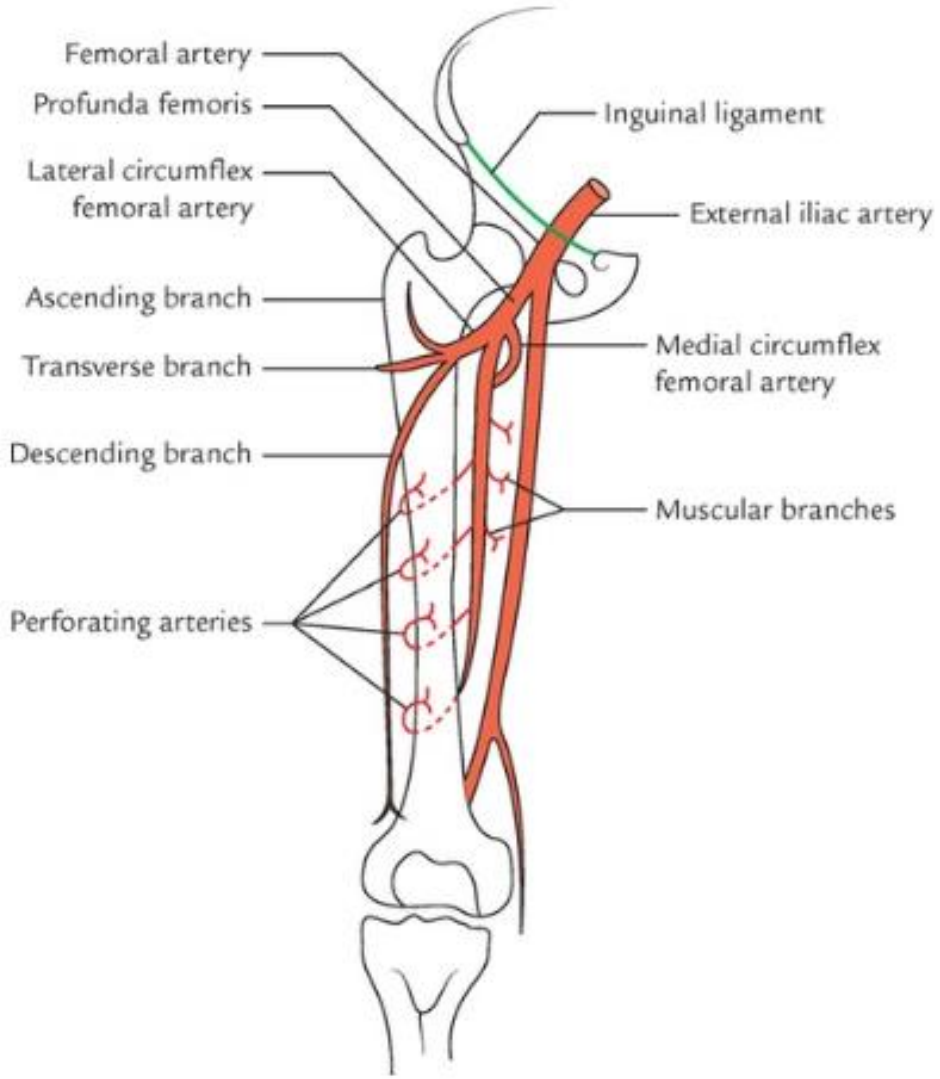


A common variant (33%) of obturator artery communication

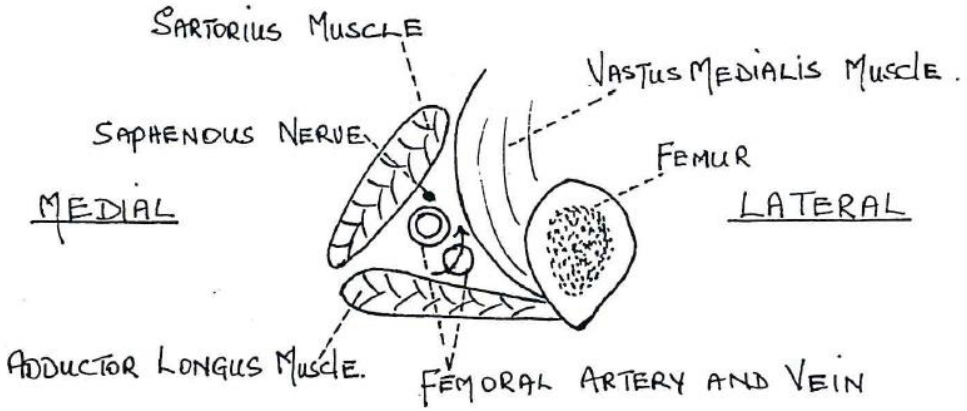


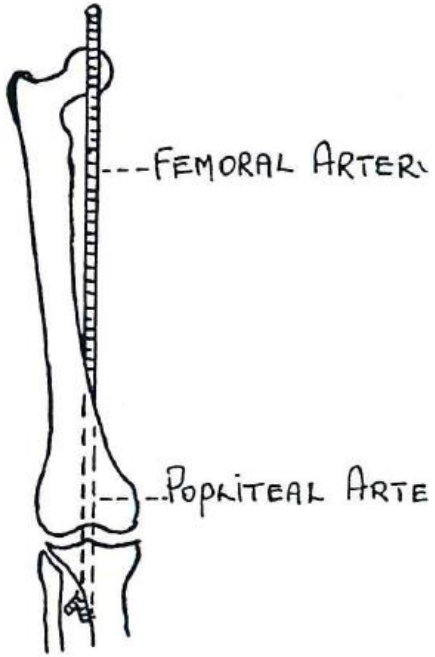
Corona mortis: crown of death
(could be damaged in femoral hernia repair)

Profunda femoris can be injured in femoral shaft fracture or by orthopaedic screws



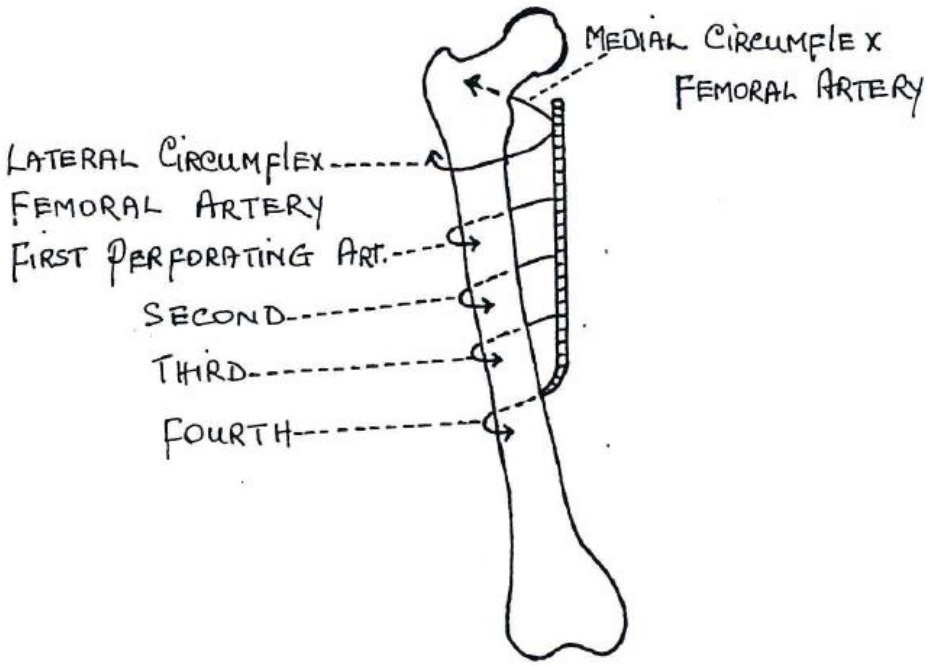
Subsartorial (adductor) canal (Hunter's canal)





Coakley Diagram

Blood supply to medial compartment



Coakley Diagram

Q. All the muscles of the adductor compartments of the thigh are Inserted into femur except:

A. Adductor magnus
B. Adductor longus
C. Gracilis
D. Adductor brevis
E. Pectineus

The NEW ENGLAND JOURNAL of MEDICINE

Incarcerated Obturator Hernia

Weisheng Yang, M.D., and Liang Peng, M.D.

rare but interesting



An 84-year-old woman with a history of six vaginal deliveries presented to the gastroenterology department with a 12-hour history of left lower abdominal pain, nausea, and vomiting. Her body-mass index (the weight in kilograms divided by the square of the height in meters) was 16, indicating underweight status. On physical examination, the patient had tenderness to palpation in the left inguinal region. Computed tomography of the abdomen revealed a loop of small bowel protruding through the left obturator canal, between the pectineus muscle anteriorly and the obturator external muscle posteriorly (Panels A [axial view] and B [coronal view], asterisk). Small-bowel dilatation with fluid accumulation proximal to the herniated bowel was also observed.

* Hernia found between obturator externus and pectineus

